CLOUD COMPUTING LAB

EXPERIMENT 6

Familiarization and Usage of Cloud Services Using AWS and Open-Source Cloud Tools

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Course Code	CSE12045	**Section**	B
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1. Aim

To become familiar with different cloud services and perform basic cloud operations using Amazon Web Services (AWS), Ubuntu Linux, Oracle VirtualBox and the AWS Command Line Interface (CLI).

2. Overview of the Practical Tasks

Task	Work carried out	Evidence
1. Virtual Machine Setup	Ubuntu on VirtualBox; system update and basic utilities	Terminal output (Figures 1-3)
2. Cloud Storage	Private Amazon S3 bucket; upload, view, download and delete a file	Console and terminal screenshots (Figures 4-21)
3. Cloud Service Exploration	IAM, EC2 and VPC explored through the AWS Console	Console screenshots (Figures 22-29)
4. Monitoring and CLI	CloudWatch overview; AWS CLI commands in CloudShell	Console and CLI output (Figures 30-37)

Environment used: Ubuntu 24.04 LTS (noble) virtual machine in Oracle VirtualBox; AWS Management Console, Region Asia Pacific (Sydney) `ap-southeast-2`; AWS CloudShell with AWS CLI 2.39.6.

3. Task 1: Virtual Machine Setup (Ubuntu on VirtualBox)

Oracle VirtualBox was used to host an Ubuntu Linux virtual machine. The terminal prompt in the screenshots below (`sk-naser-hussain@sk-naser-hussain-VirtualBox`) shows the working Ubuntu guest, and the package sources point to the Ubuntu *noble* (24.04 LTS) repositories.

Step 3: Update Ubuntu

`sudo apt update` refreshes the package index from the Ubuntu archive and security mirrors, and `sudo apt upgrade -y` installs the newer versions of the installed packages without asking for confirmation.

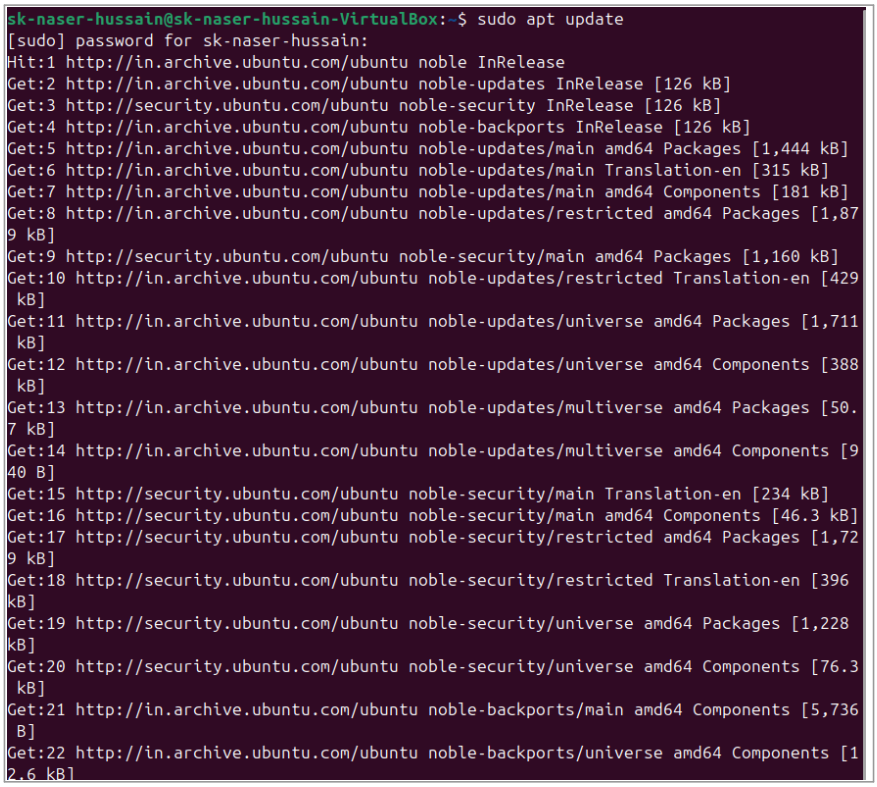


Figure 1: Running sudo apt update to refresh the package lists

The upgrade installed 32 packages (about 145 MB of archives) and reported 3 packages not upgraded; three packages (distro-info, python3-distro-info, snapd) were deferred due to phased updates.

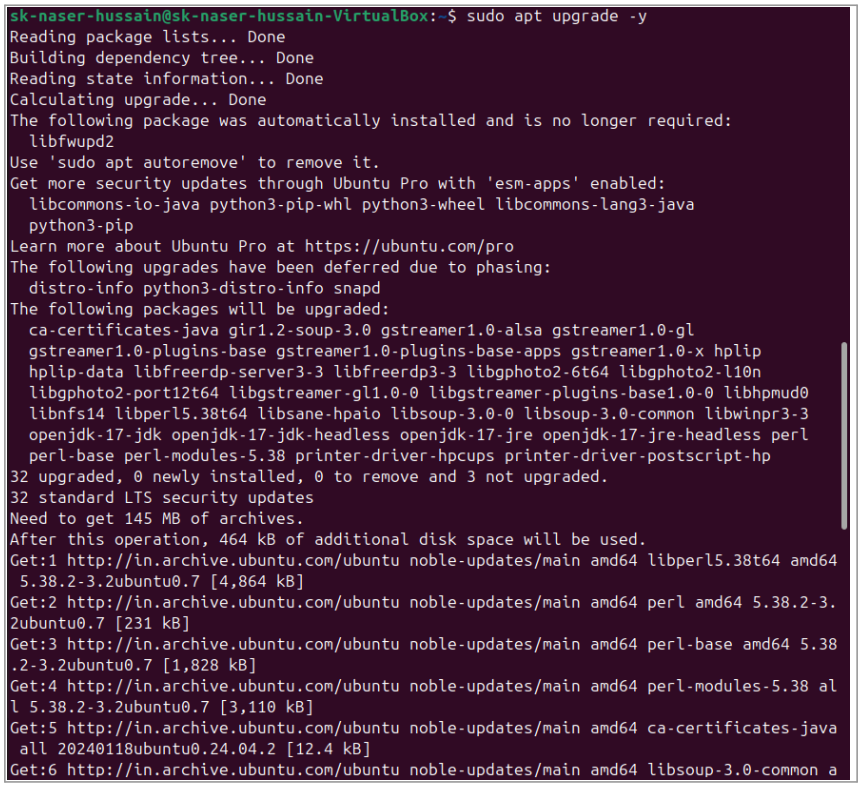


Figure 2: Running sudo apt upgrade -y (32 upgraded, 0 newly installed, 0 to remove, 3 not upgraded)

Step 4: Install Basic Utilities

The utilities *curl, wget, git, nano* and *tree* were installed with `sudo apt install -y curl wget git nano tree`. wget, git and nano were already at their newest versions, so only curl and tree were newly installed.

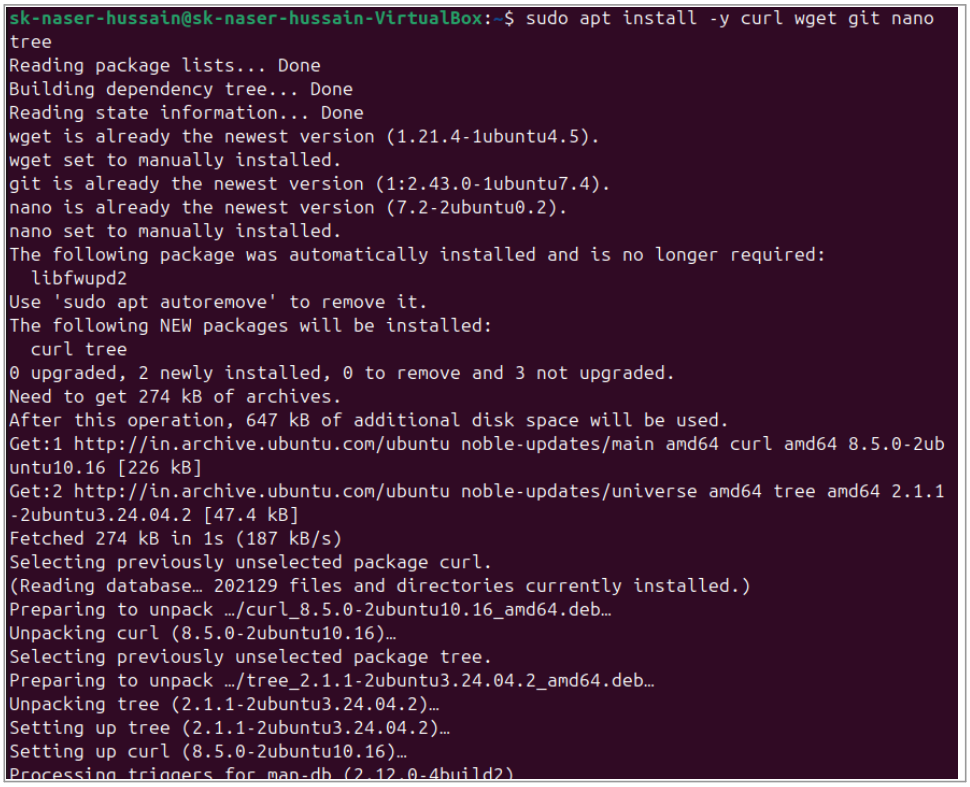


Figure 3: Installing basic utilities (curl and tree installed; wget, git and nano already up to date)

4. Task 2A: Introduction to the AWS Management Console

Amazon Web Services (AWS) is a cloud computing platform that provides computing, storage, databases, networking, security and monitoring services, allowing users to create and manage cloud resources without maintaining physical infrastructure. After signing in to the AWS Management Console with an authorised account, the search bar at the top was used to locate services; searching for *s3* lists Amazon S3 under Services.



Figure 4: AWS Management Console search bar locating the S3 service

5. Task 2B: Cloud Storage Using Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service used to store and retrieve data over the internet. Data is stored as objects inside buckets, and bucket names must be globally unique.

Step 1 and 2: Open S3 and create a bucket

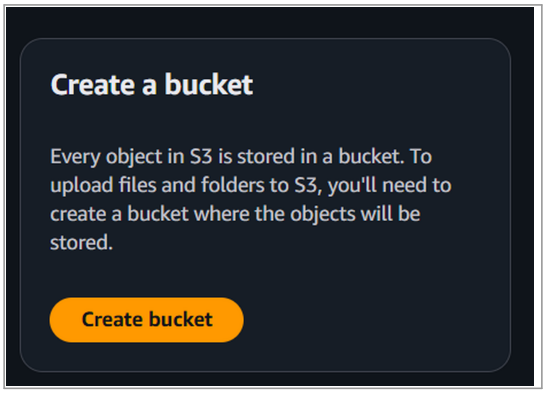


Figure 5: Create bucket option in the Amazon S3 console

The bucket was configured as follows:

Setting	Value chosen
AWS Region	Asia Pacific (Sydney) ap-southeast-2
Bucket type / namespace	General purpose / Global namespace
Bucket name	`cse-cloud-lab6-naser-2026`
Object Ownership	ACLs disabled (bucket owner enforced)
Block Public Access	Block *all* public access enabled (all four options)
Bucket Versioning	Disabled

Setting	Value chosen
Default encryption	Server-side encryption with Amazon S3 managed keys (SSE-S3); Bucket Key disabled

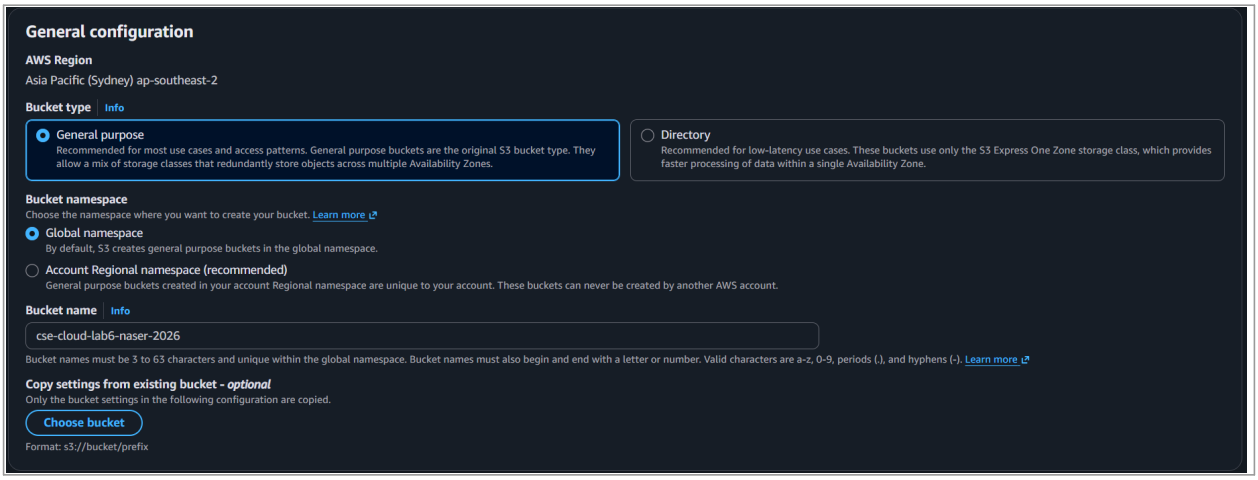


Figure 6: General configuration: Region, bucket type, namespace and globally unique bucket name

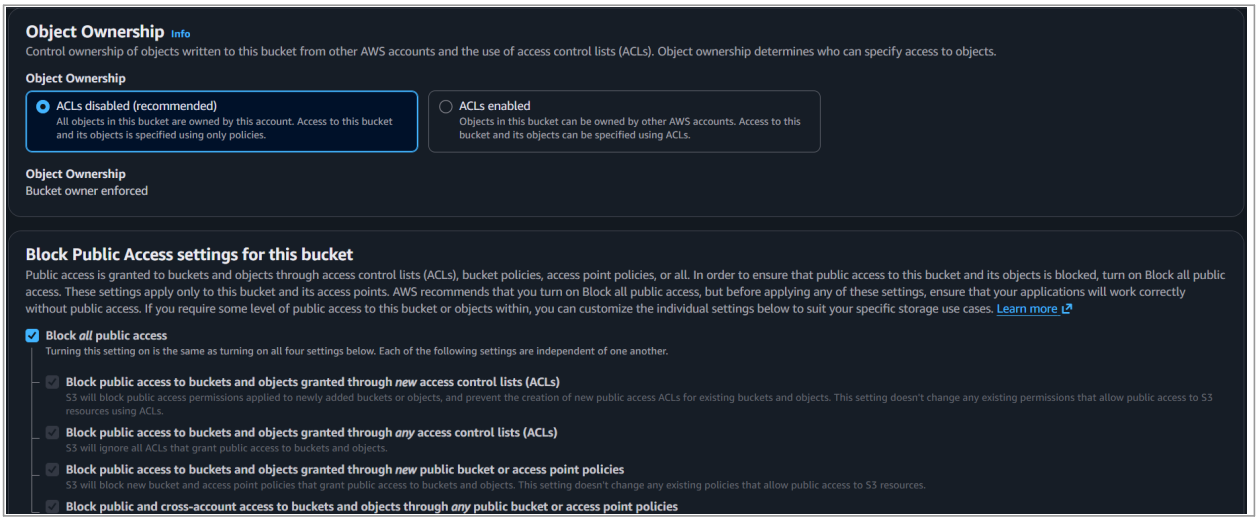


Figure 7: Object Ownership and Block Public Access settings (public access blocked)



Figure 8: Bucket Versioning left disabled

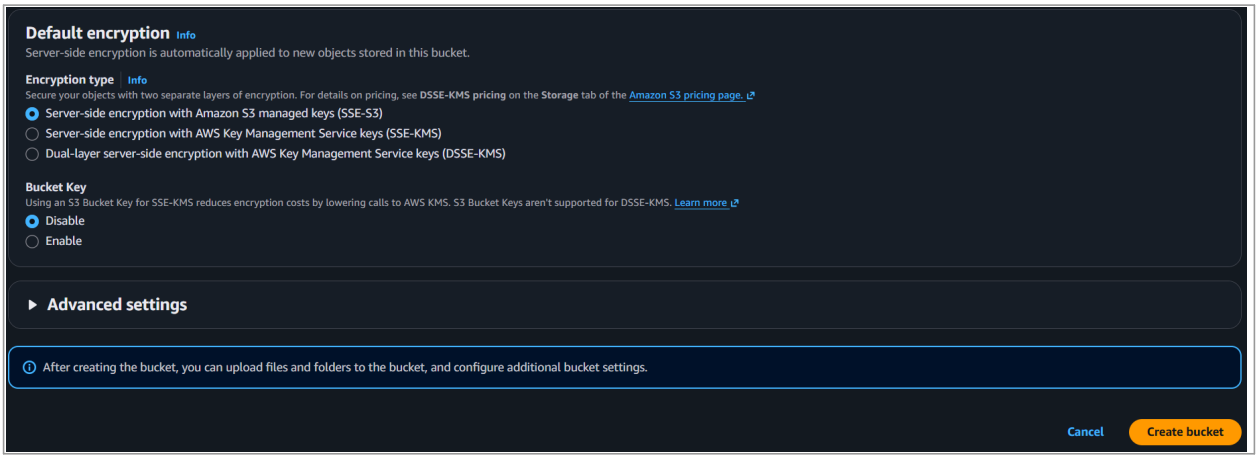


Figure 9: Default encryption (SSE-S3) and the Create bucket button

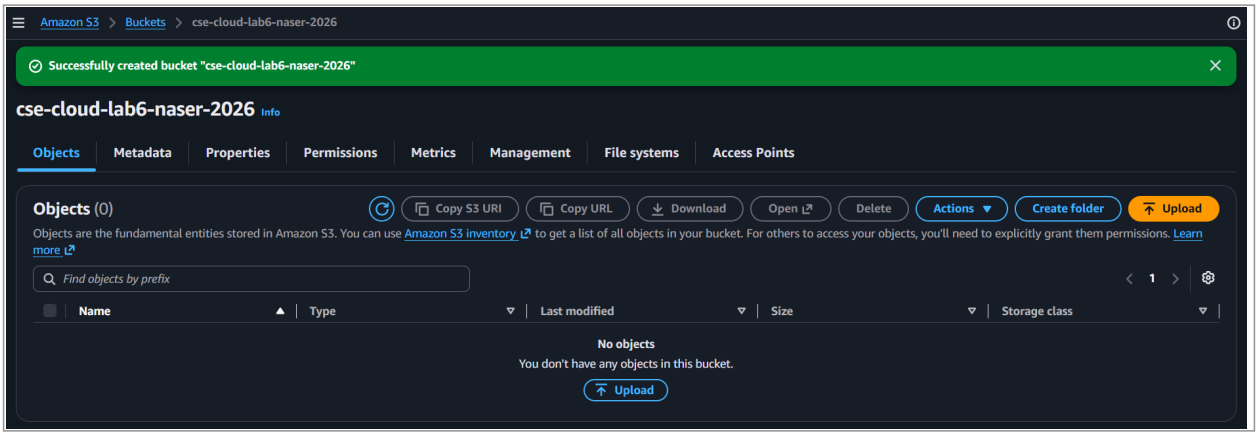


Figure 10: Bucket created successfully; the Objects tab is empty

Step 3: Create a sample file in Ubuntu

A directory `cloud-lab6` was created, entered, and a text file was written and displayed. *mkdir* creates a directory, *cd* changes the current directory, *echo* writes text into a file and *cat* displays its contents. The first *cd* attempt contained a typing error (*clou-lab6*), which produced a "No such file or directory" message; it was corrected on the next command.

```
mkdir cloud-lab6
cd cloud-lab6
echo "Cloud Computing Lab Experiment 6" > lab-note.txt
cat lab-note.txt
```

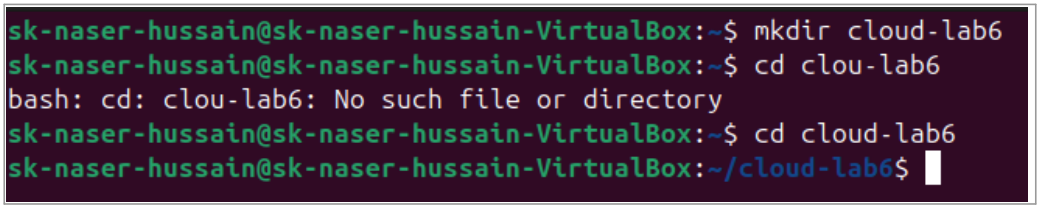


Figure 11: Creating and entering the cloud-lab6 directory

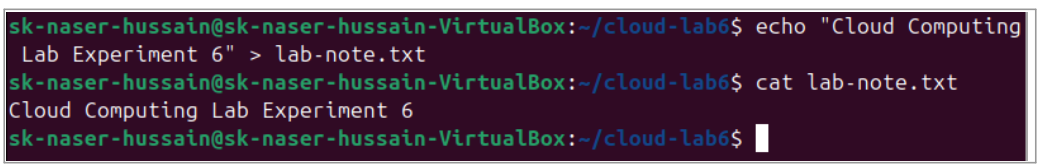


Figure 12: Creating lab-note.txt with echo and displaying it with cat

Step 4: Upload the file to S3

Inside the bucket, *Upload* then *Add files* was used to select `lab-note.txt` (33 bytes: 32 characters plus the newline added by echo). The destination was the bucket `s3://cse-cloud-lab6-naser-2026`.

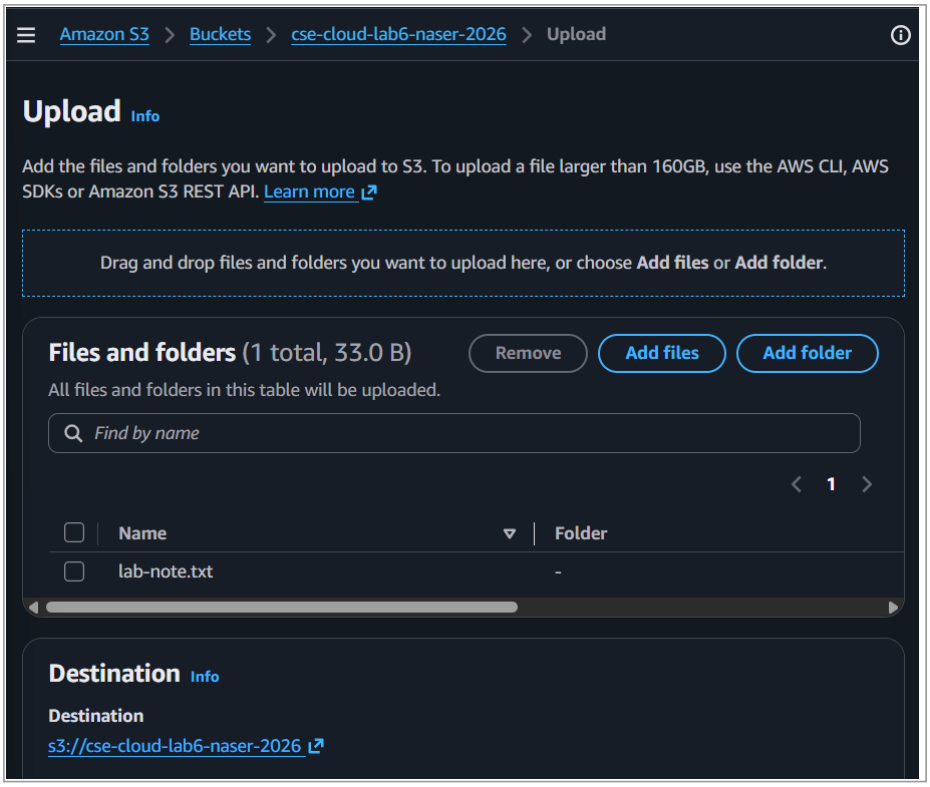


Figure 13: Upload page with lab-note.txt (33.0 B) selected and the bucket as destination

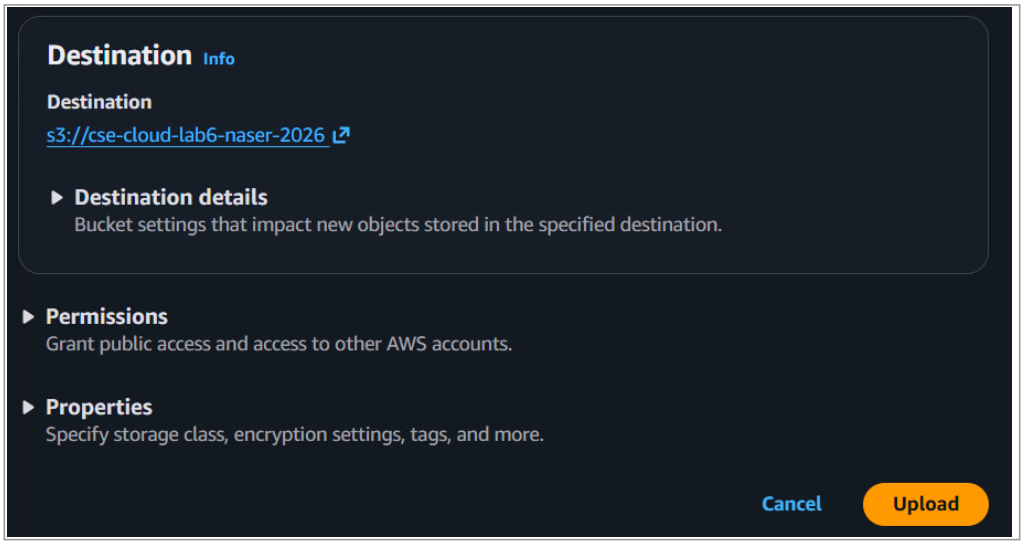


Figure 14: Destination, Permissions and Properties sections with the Upload button

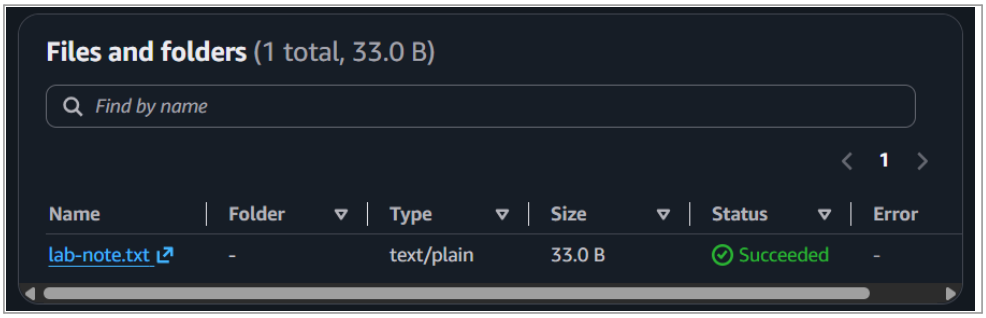


Figure 15: Upload status: lab-note.txt (text/plain, 33.0 B) succeeded

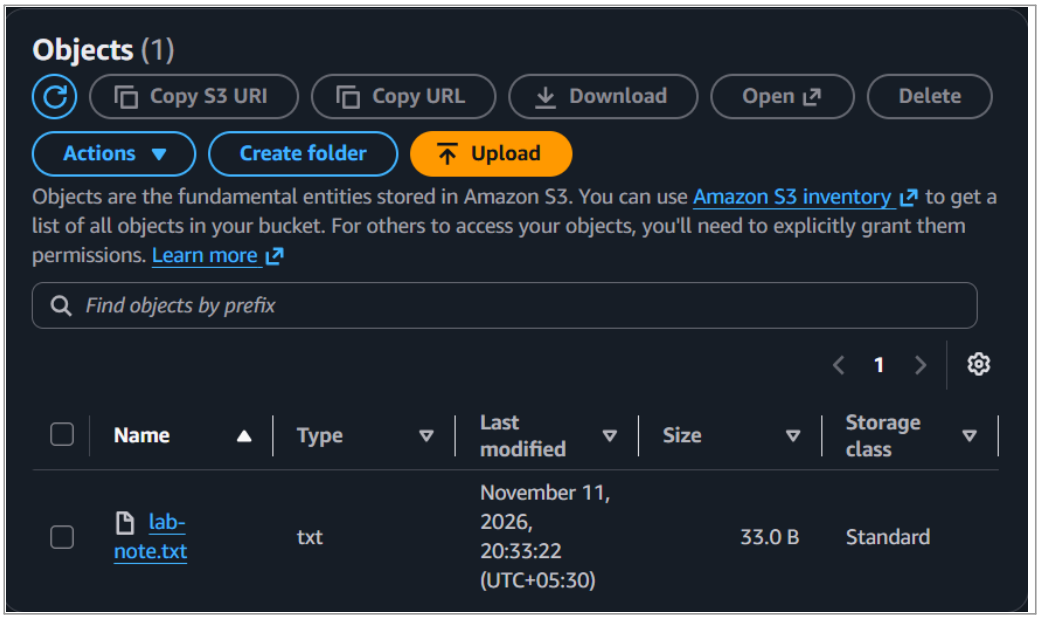


Figure 16: The object appears in the bucket (storage class Standard)

Step 5: Download and verify the file

The uploaded object was selected and downloaded using the *Download* button. The browser download history shows *lab-note.txt, 33 B, Done*, and opening the file shows the same text as the original, with 33 characters, confirming that the contents match.

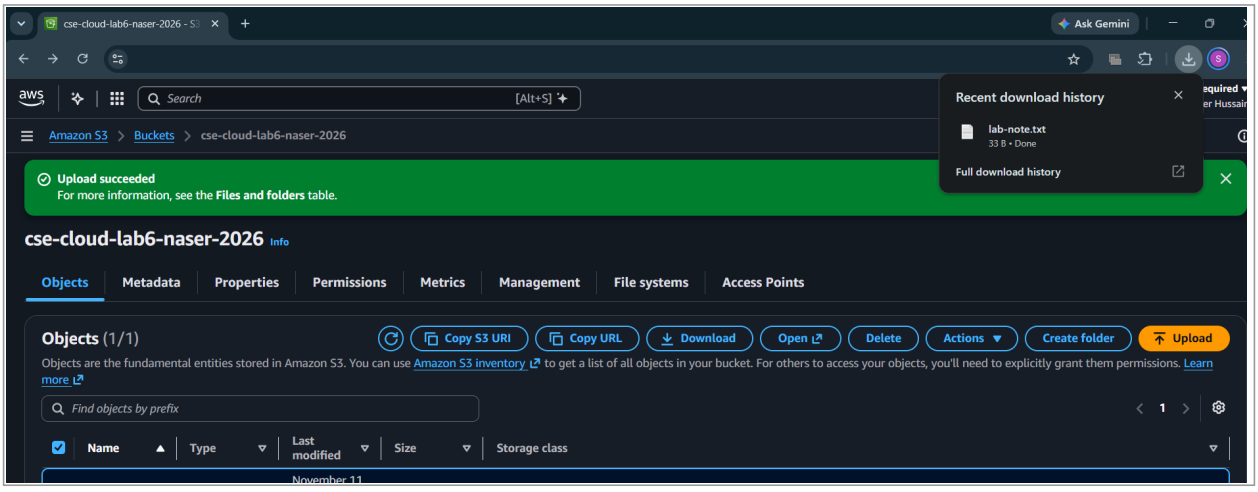


Figure 17: Upload succeeded message and the downloaded lab-note.txt (33 B) in the browser download history

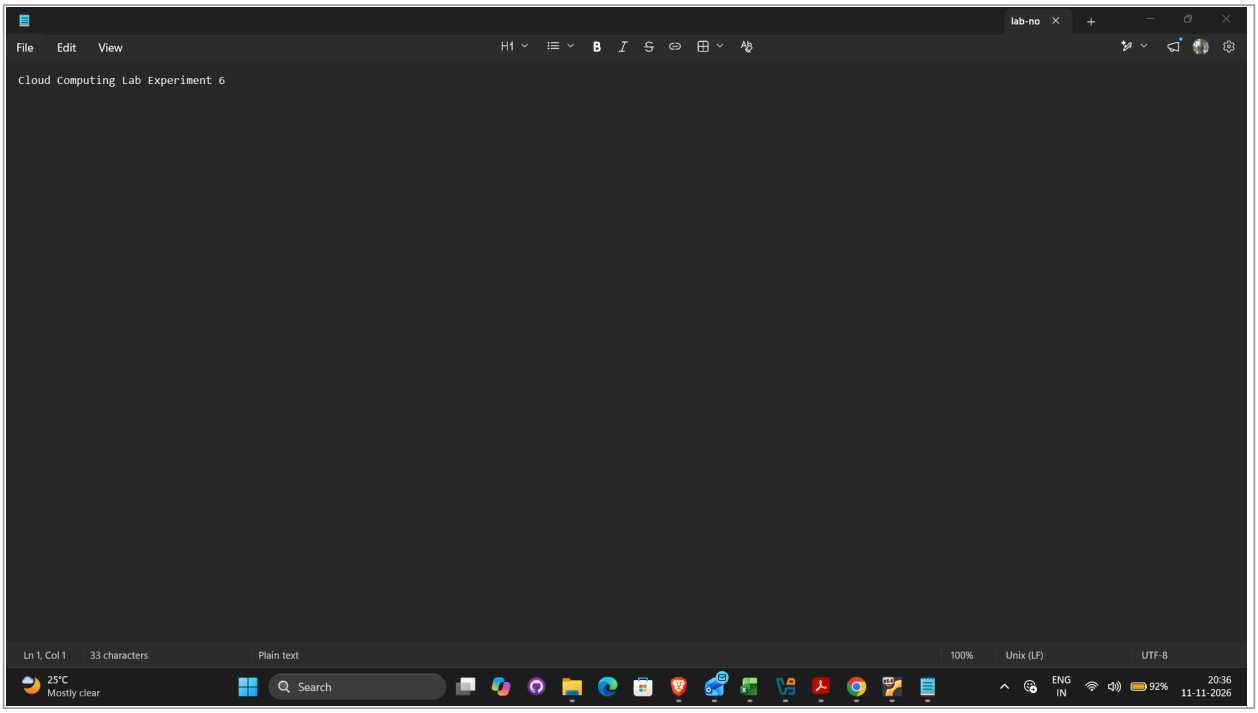


Figure 18: Downloaded lab-note.txt opened in a text editor: contents match the original file

Step 6: Delete the test object and the bucket

To avoid leaving unused resources, the object was deleted first and then the (now empty) bucket was deleted after typing the bucket name to confirm.

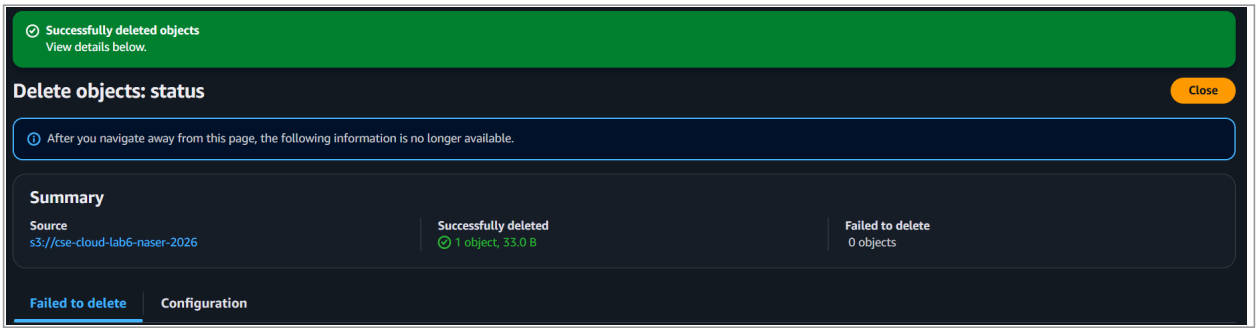


Figure 19: Delete objects status: 1 object (33.0 B) successfully deleted, 0 failed

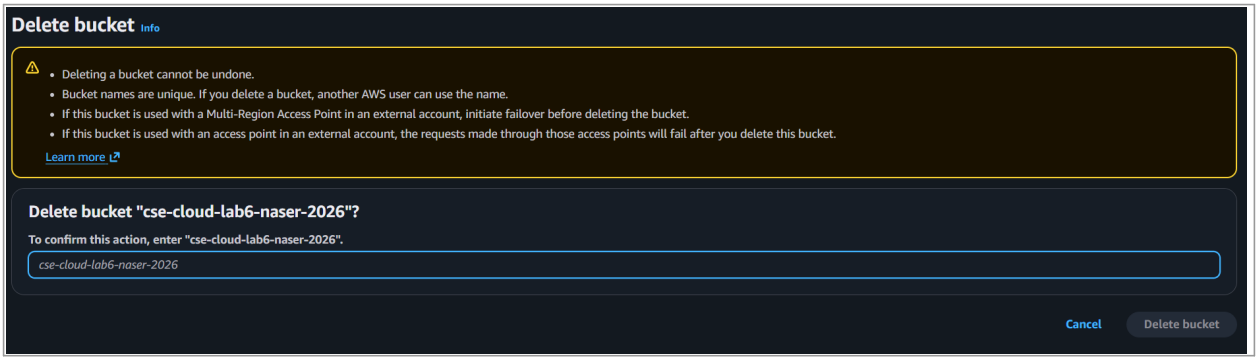


Figure 20: Delete bucket confirmation prompt (bucket name must be typed to confirm)

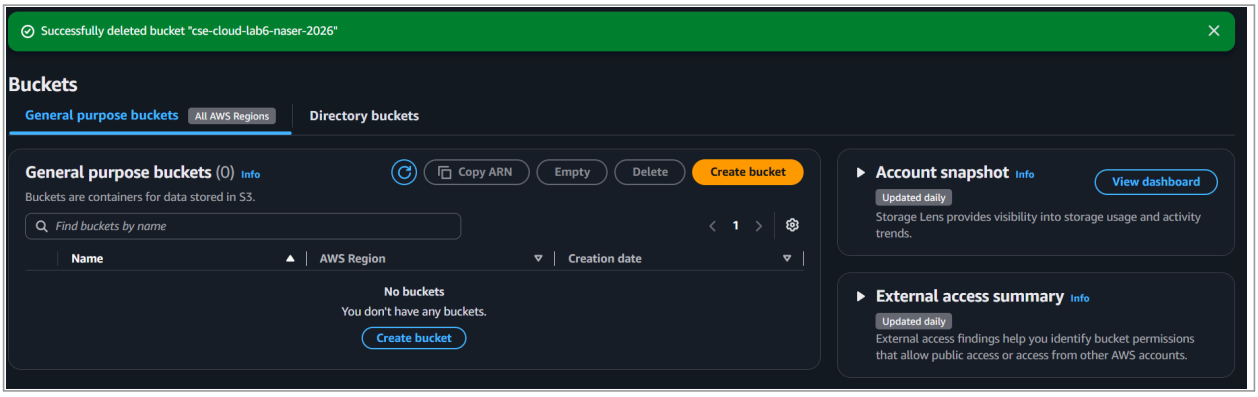


Figure 21: Bucket successfully deleted; the account now lists no buckets

Result: A private S3 bucket was created, a sample file was uploaded, downloaded and verified, and then the object and bucket were deleted.

6. Task 3A: Identity and Access Management Using AWS IAM

AWS Identity and Access Management (IAM) controls who can access AWS resources and what actions they can perform. The IAM dashboard was opened through the console search and its Roles, Policies, Users and Root access management sections were explored as an observation activity.

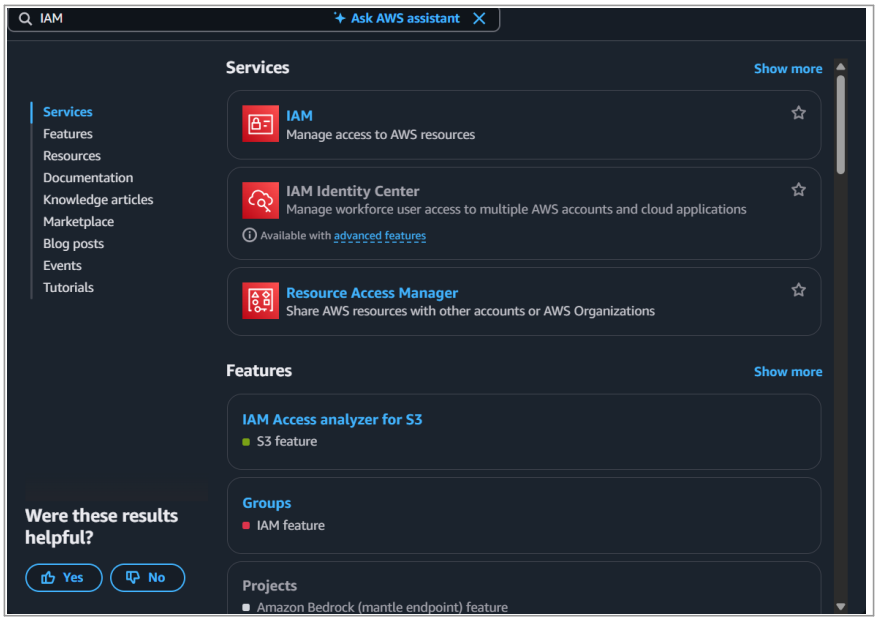


Figure 22: Searching for IAM in the AWS Console

Roles: A role is an identity with specific permissions and short-lived credentials that trusted entities can assume. The account shows 10 roles, including *AccountFullAccessRole* (which is the role the CloudShell session later reports in Task 4B) and several AWS service-linked roles.

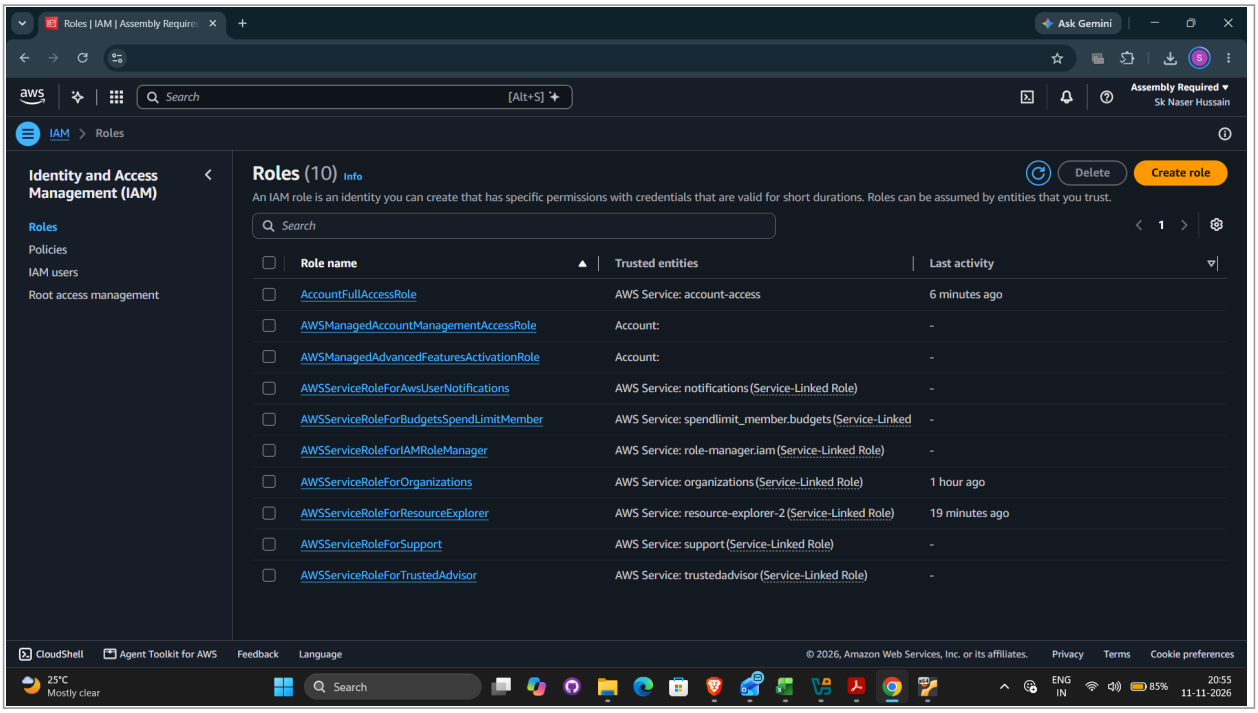


Figure 23: IAM Roles page listing 10 roles and their trusted entities (account numbers hidden)

Policies: A policy is an object that defines permissions. The console lists 1601 policies, mostly AWS managed (for example *AdministratorAccess*).

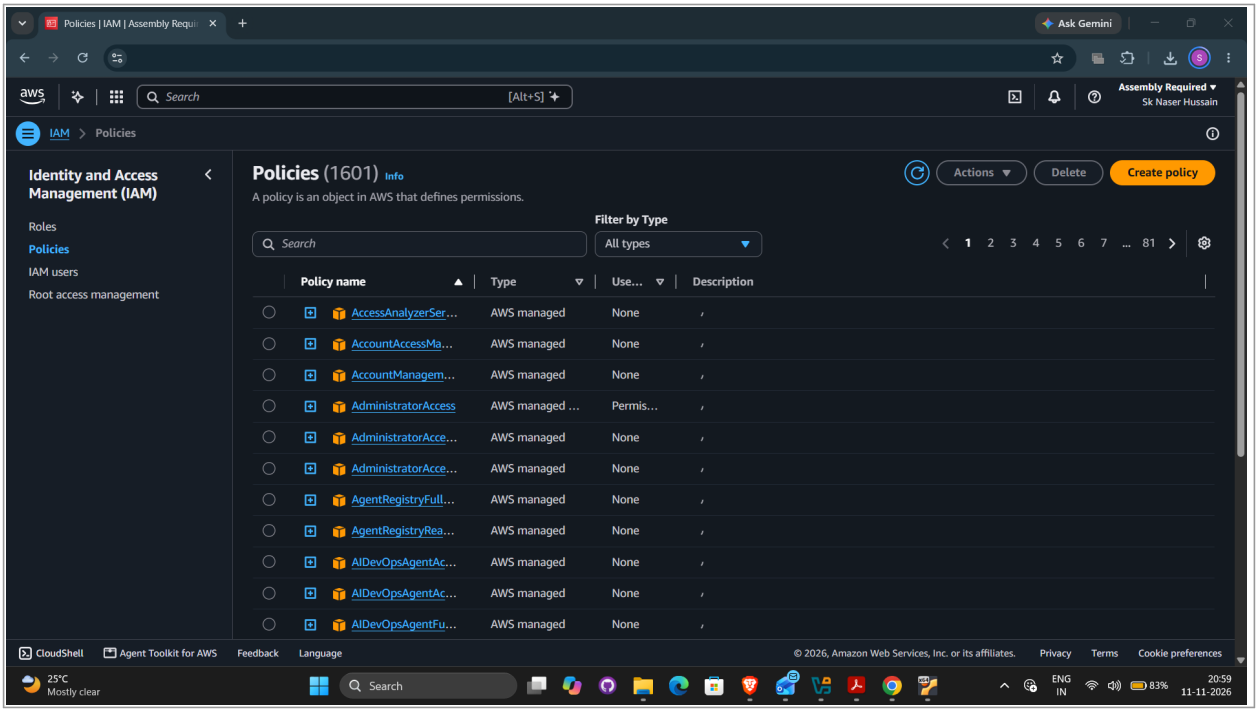


Figure 24: IAM Policies page (1601 policies, mostly AWS managed)

Users: The account has no IAM users (0). The console notes that IAM users should only be used for specific programmatic access, and human access should be given through the recommended federated route.

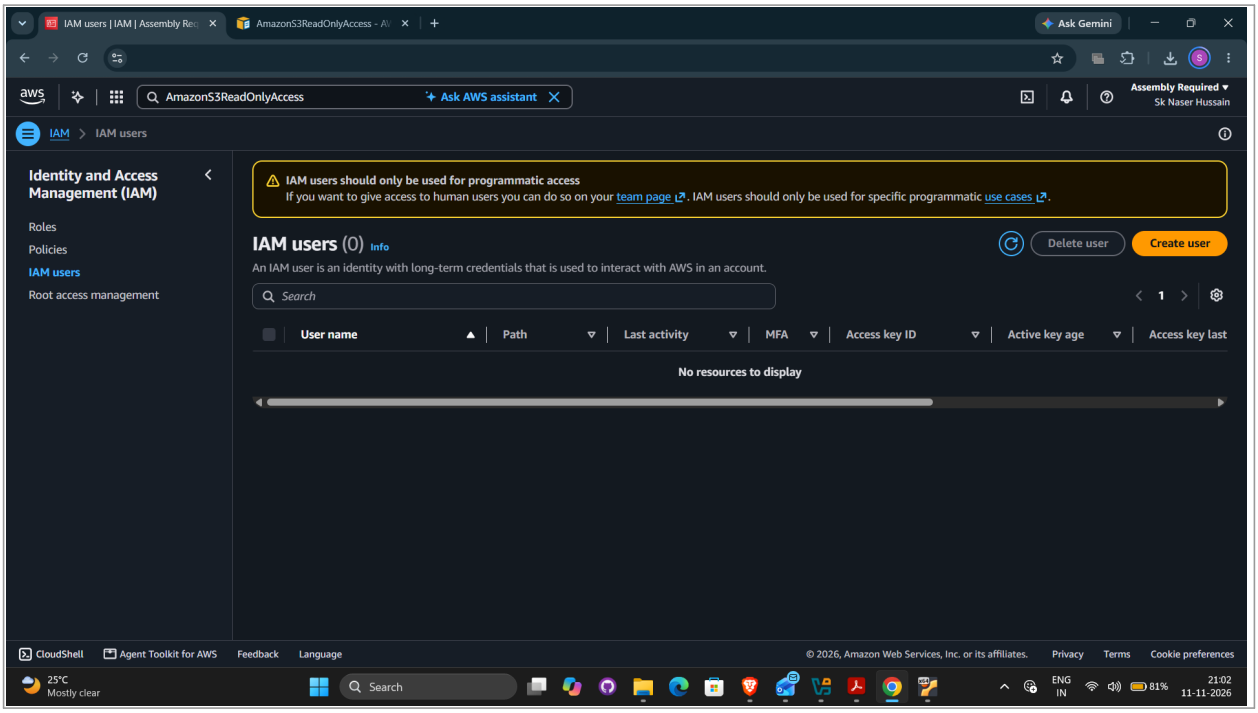


Figure 25: IAM users page showing no IAM users

Root access management: This page lets an administrator launch a privileged session for a curated set of root-only actions (deleting a misconfigured S3 bucket policy or SQS queue policy) without using root credentials.

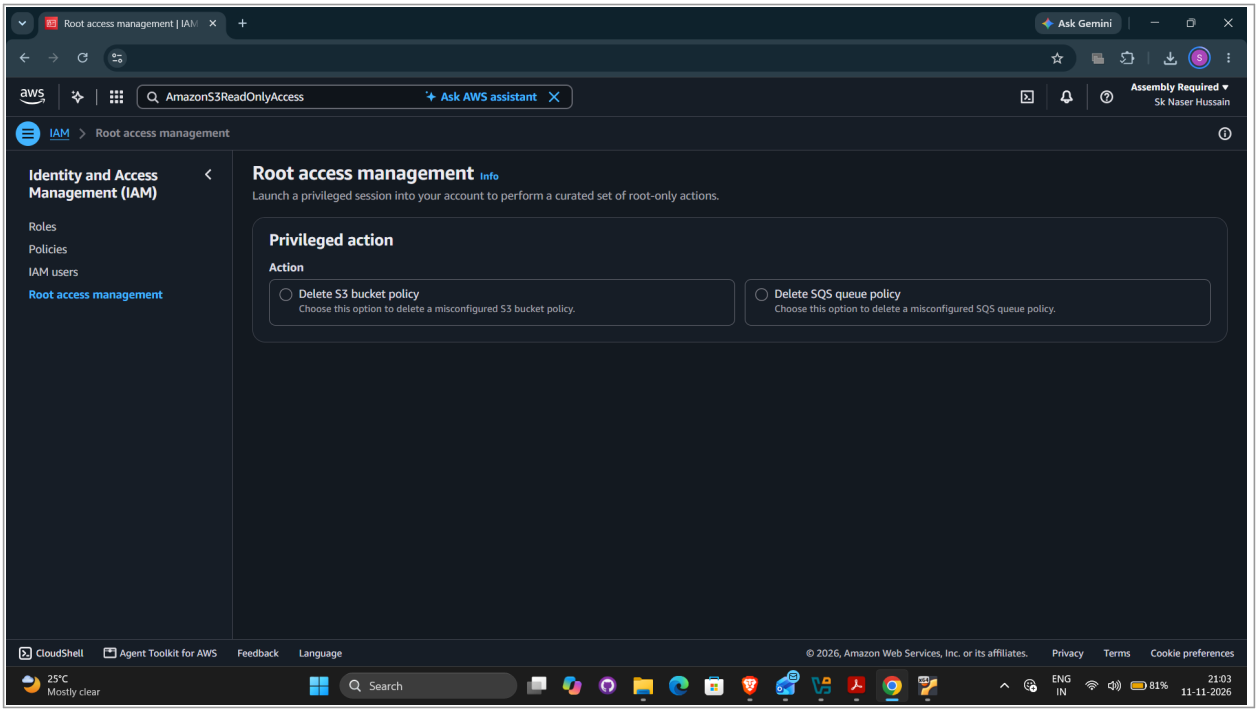


Figure 26: Root access management page with the available privileged actions

IAM policy structure

An IAM policy is a JSON document. The illustrative policy below allows listing the contents of one bucket; it does not allow uploading or downloading objects.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [ "s3:ListBucket" ],
      "Resource": "arn:aws:s3:::example-bucket"
    }
  ]
}
```

- **Version** specifies the policy language version.
- **Effect** states whether the action is allowed or denied.
- **Action** names the AWS operation (here, listing a bucket).
- **Resource** identifies the AWS resource the permission applies to.

Principle of least privilege: a user or role should receive only the minimum permissions needed to perform its assigned tasks. No users, access keys or policies were created or changed, in line with the safety note of the manual.

7. Task 3B: Virtual Machine Provisioning Using Amazon EC2

Amazon Elastic Compute Cloud (EC2) provides virtual servers; an EC2 instance behaves like a virtual machine hosted on AWS infrastructure. The EC2 dashboard for the Sydney Region shows 0 running instances, 0 volumes, 0 key pairs and 1 security group, so no billable compute or storage resources were present.

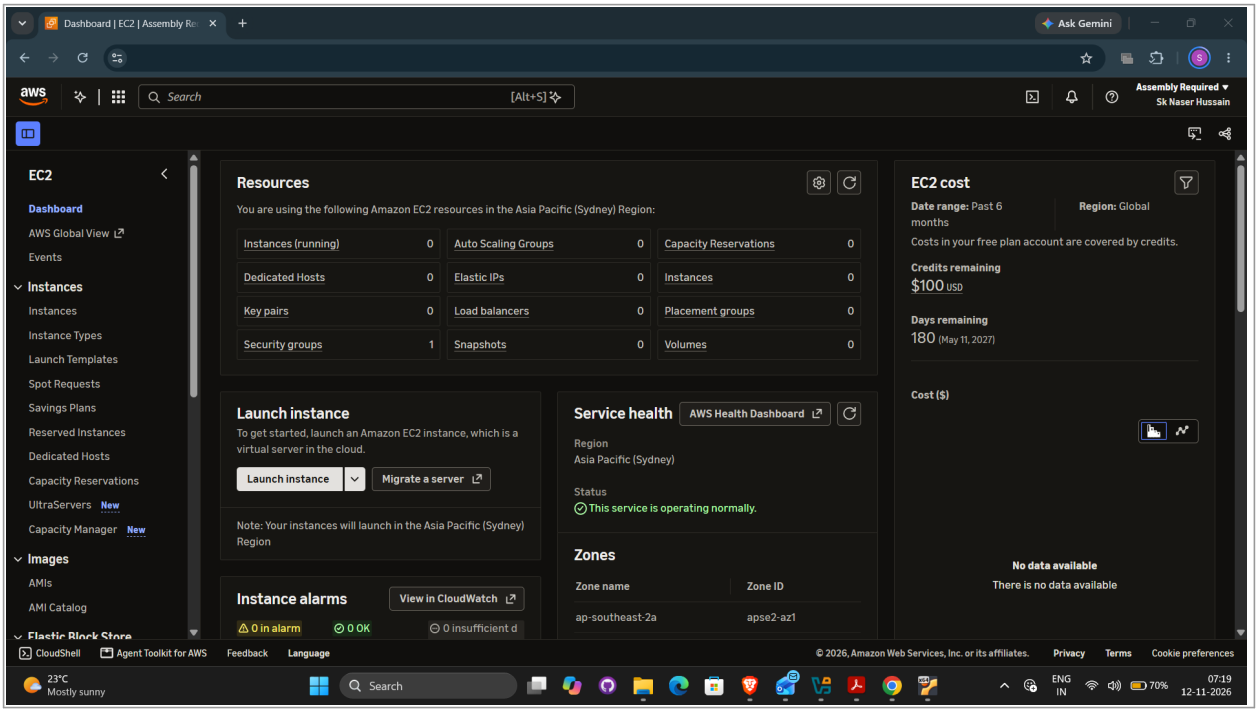


Figure 27: EC2 dashboard (Asia Pacific, Sydney): resource summary and Launch instance option

The *Launch instance* screen was used to explore the launch process: the Name and tags field, the Amazon Machine Image (AMI) catalogue (Amazon Linux, macOS, Ubuntu, Windows, Red Hat, SUSE Linux, Debian), the instance type, key pair, network and security-group settings and storage. The default summary shows the Amazon Linux 2023 AMI, instance type t3.micro, a new security group and one 8 GiB volume. As allowed by the manual, the experiment stopped at the review stage and no instance was launched, since charges depend on instance type, Region, operating system, storage and account eligibility.

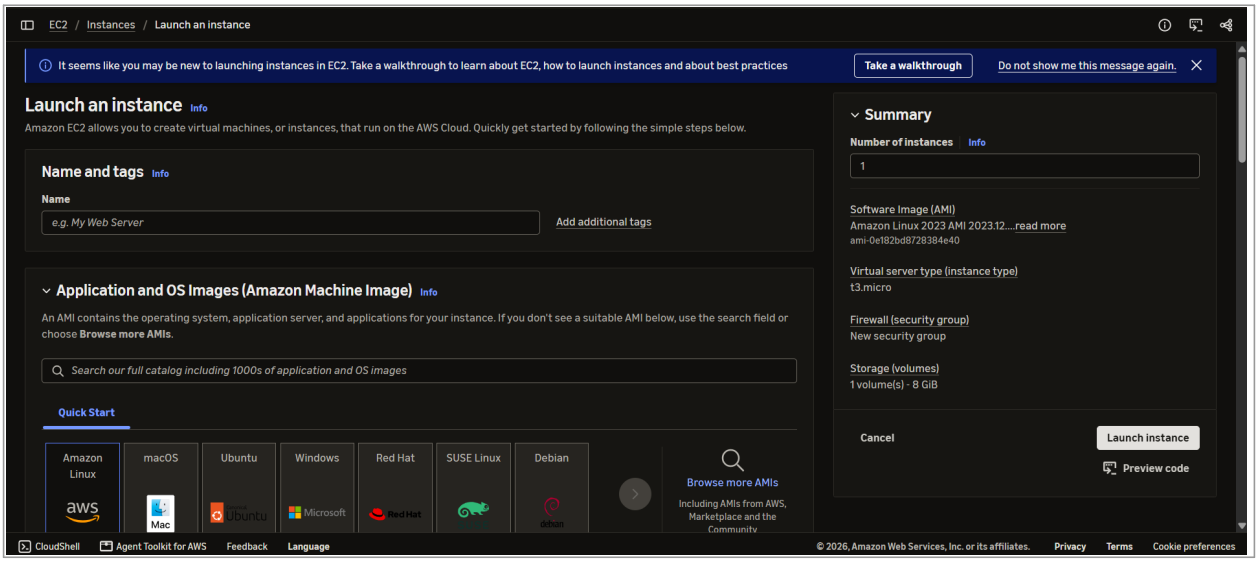


Figure 28: Launch an instance page: AMI selection and the configuration summary (not launched)

8. Task 3C: Virtual Networking Using Amazon VPC

Amazon Virtual Private Cloud (VPC) provides a logically isolated virtual network in AWS, with control over IP address ranges, subnets, routing and network security. The VPC dashboard for Sydney shows the default networking components present in the account:

Component	Count (Sydney)
VPCs	1
Subnets	3
Route Tables	1
Internet Gateways	1
Network ACLs	1
Security Groups	1
NAT Gateways / Peering connections / Endpoint services	0

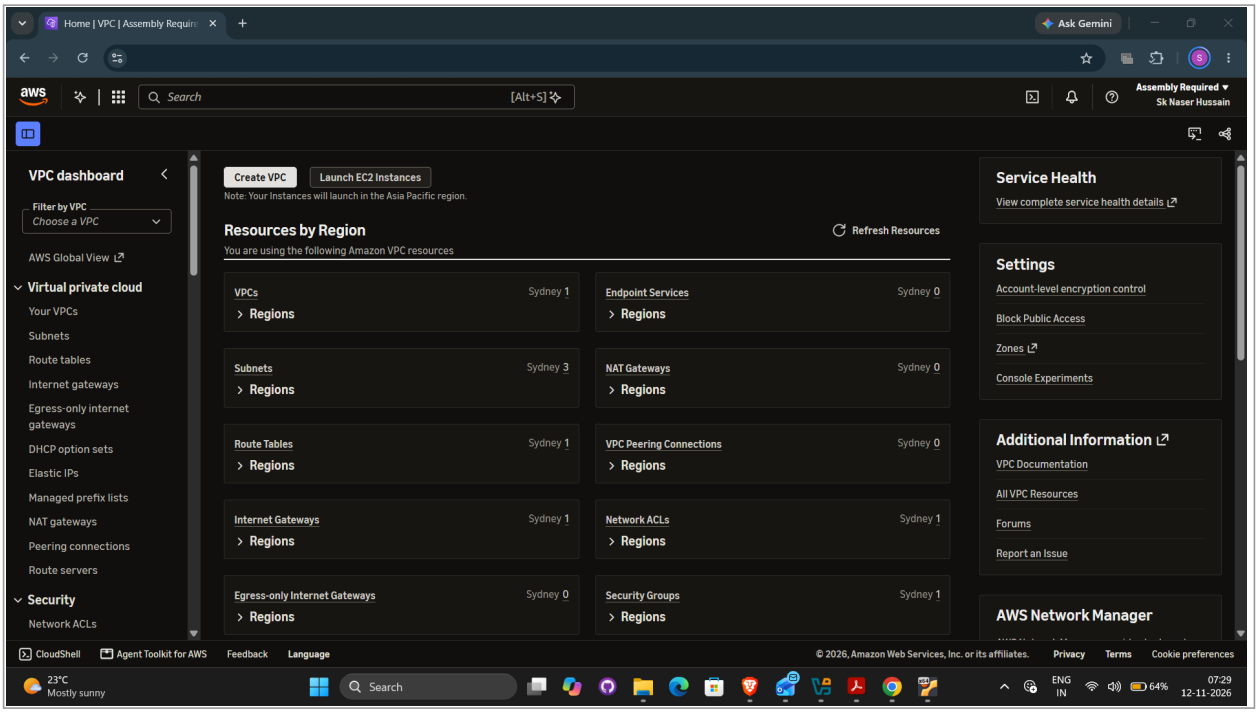


Figure 29: VPC dashboard: resources by region (VPC, subnets, route table, internet gateway, network ACL, security group)

A typical architecture contains a VPC, a public subnet, a private subnet, an internet gateway, route tables and security groups. A public subnet gives internet connectivity when its route table sends traffic to an internet gateway and its security rules allow it, whereas a private subnet generally has no direct route to an internet gateway. Security groups act as virtual firewalls with inbound and outbound rules. The default VPC resources were only observed and were not modified or deleted.

9. Task 4A: Cloud Monitoring Using Amazon CloudWatch

Amazon CloudWatch is a monitoring and observability service that collects and displays metrics, logs and other monitoring information for AWS resources and applications. The CloudWatch overview shows 0 alarms in any state and indicates that no alarms, metrics or default dashboard exist yet in the account.

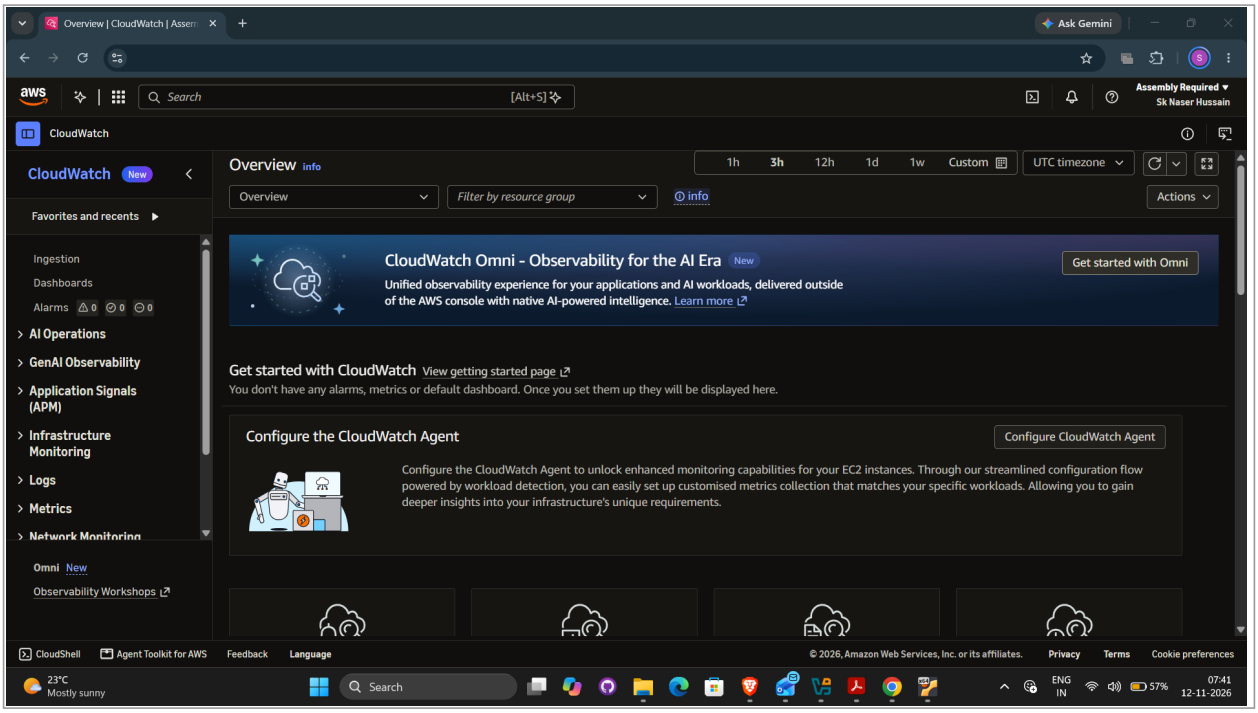


Figure 30: CloudWatch overview: no alarms (0 in alarm, 0 OK, 0 insufficient data)

- **Metrics** are organised by service and by dimension; for example, an EC2 instance provides CPU utilisation metrics.
- **Alarms** watch a selected metric against a threshold and change state (OK, In alarm, Insufficient data) when the threshold condition is met. For example, an alarm can notify an administrator when CPU utilisation exceeds a set percentage for a defined period.

No dashboards, log groups or alarms were created, to avoid unnecessary charges.

10. Task 4B: Basic AWS Operations Using the AWS CLI

The AWS CLI lets users manage AWS services through terminal commands instead of the console. AWS CloudShell, a browser-based terminal that runs in the authenticated account, was used so that no access keys had to be created or stored.

Step 1: Open CloudShell

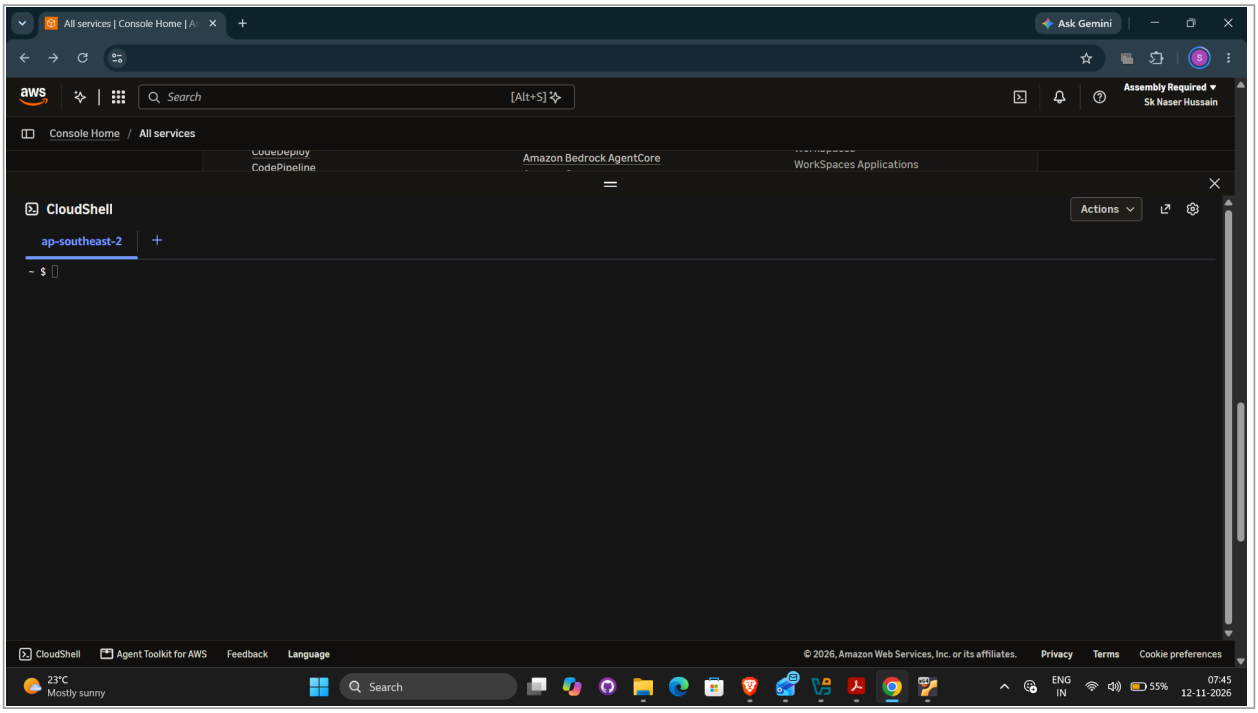


Figure 31: AWS CloudShell opened in the ap-southeast-2 Region

Step 2: Verify the AWS CLI installation

```
aws --version
```



Figure 32: AWS CLI version 2.39.6 running on Amazon Linux in CloudShell

Step 3: Verify the AWS identity

`aws sts get-caller-identity` returns the user ID, account number and ARN of the credentials in use. The session is using the assumed role *AccountFullAccessRole*, i.e. temporary credentials. The identifying values are hidden in this report as advised by the manual.



Figure 33: Caller identity output (values redacted)

Step 4: List available S3 buckets

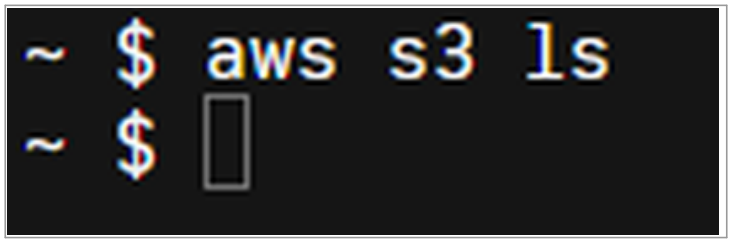


Figure 34: aws s3 ls returns no buckets because the console-created bucket had already been deleted

Steps 5 to 10: Upload, list, download, verify and delete using the CLI

A local file was created first. Because the console bucket from Task 2B had been deleted, a new bucket with the same name was created from the CLI with `aws s3 mb` before uploading.

```
mkdir cloud-lab6-cli
cd cloud-lab6-cli
echo "AWS CLI Cloud Storage Test" > sample.txt
```

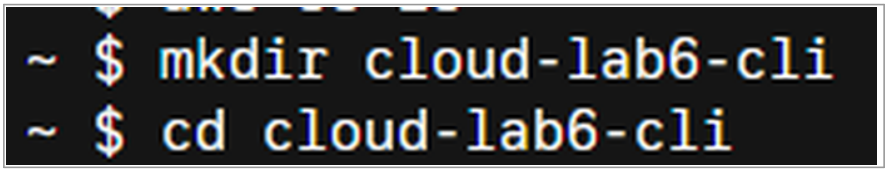


Figure 35: Creating the cloud-lab6-cli directory



Figure 36: Creating and displaying sample.txt

The remaining commands, run in order, were:

Command	Purpose
`aws s3 mb s3://cse-cloud-lab6-naser-2026 --region ap-southeast-2`	Create the bucket (make_bucket)
`aws s3 cp sample.txt s3://cse-cloud-lab6-naser-2026/`	Upload the file to S3
`aws s3 ls s3://cse-cloud-lab6-naser-2026`	List objects in the bucket (sample.txt, 27 bytes)
`aws s3 cp s3://cse-cloud-lab6-naser-2026/sample.txt downloaded.txt`	Download the object
`cat downloaded.txt`	Verify the downloaded content
`aws s3 rm s3://cse-cloud-lab6-naser-2026/sample.txt`	Delete the test object
`aws s3 rb s3://cse-cloud-lab6-naser-2026`	Remove the (empty) test bucket
`aws s3 ls`	Confirm that no buckets remain

```
cloud-lab6-cli $ aws s3 mb s3://cse-cloud-lab6-naser-2026 --region ap-southeast-2
make_bucket: cse-cloud-lab6-naser-2026
cloud-lab6-cli $ ^C
cloud-lab6-cli $ aws s3 cp sample.txt s3://cse-cloud-lab6-naser-2026/
upload: ./sample.txt to s3://cse-cloud-lab6-naser-2026/sample.txt
cloud-lab6-cli $ aws s3 ls s3://cse-cloud-lab6-naser-2026
2026-11-12 02:26:18         27 sample.txt
cloud-lab6-cli $ aws s3 cp s3://cse-cloud-lab6-naser-2026/sample.txt downloaded.txt
download: s3://cse-cloud-lab6-naser-2026/sample.txt to ./downloaded.txt
cloud-lab6-cli $ cat downloaded.txt
AWS CLI Cloud Storage Test
cloud-lab6-cli $ aws s3 rm s3://cse-cloud-lab6-naser-2026/sample.txt
delete: s3://cse-cloud-lab6-naser-2026/sample.txt
cloud-lab6-cli $ aws  s3 rb s3://cse-cloud-lab6-naser-2026
remove_bucket: cse-cloud-lab6-naser-2026
cloud-lab6-cli $ aws s3 ls
cloud-lab6-cli $
```

Figure 37: AWS CLI session: bucket creation, upload, listing, download, verification, deletion and cleanup

The downloaded file displays *AWS CLI Cloud Storage Test*, which matches the original file. The final *aws s3 ls* prints nothing, confirming that the test object and bucket were removed. (The stray *^C* in the session is an accidental keypress and had no effect on the commands.)

Result: The content of the downloaded file matched the original file, and all test resources were cleaned up.

11. Precautions and AWS Cost Management

AWS Free Tier does not mean every service and configuration is free, so the following precautions from the manual were observed:

Precaution	How it was followed in this experiment
Keep S3 buckets private	Block all public access was enabled; default SSE-S3 encryption kept on (Figure 7, Figure 9).
No unnecessary IAM changes	IAM was observation only; no users, access keys or policies were created or changed (Figures 22-26).
No unapproved EC2 launch	The launch process was explored only up to the configuration summary; the dashboard shows 0 running instances (Figures 27-28).
Do not modify default VPC resources	The VPC dashboard was only viewed (Figure 29).
Avoid unnecessary monitoring resources	No CloudWatch dashboards, log groups or alarms were created (Figure 30).
Protect credentials	CloudShell used temporary role credentials; no secret keys or root credentials appear in this report, and account identifiers were hidden.
Delete test resources	The S3 object and bucket were deleted in both the console (Figures 19-21) and the CLI (Figure 37).

12. Conclusion

In this experiment an Ubuntu virtual machine was updated and prepared with basic utilities, and the core AWS services were explored. A private Amazon S3 bucket was created and a file was uploaded, downloaded and deleted through both the console and the AWS CLI. IAM, EC2, VPC and CloudWatch were examined to understand access control, virtual machines, networking and monitoring in the cloud, while following the cost and security precautions so that no resources were left running.